

EXAM PROOFS

EXAM 1

Theorem: If a function f is differentiable at a , then f is continuous at a

Proof:

Suppose f is differentiable at a . f is continuous at $a \Leftrightarrow \lim_{x \rightarrow a} f(x) = f(a)$

$$\Leftrightarrow \lim_{x \rightarrow a} f(x) - f(a) = 0$$

$$\Leftrightarrow \lim_{x \rightarrow a} (f(x) - f(a)) = 0$$

Cannot use $\Rightarrow$ because then we cannot show the bottom limit to show the top

$$\Leftrightarrow \lim_{x \rightarrow a} \left(\left(\frac{f(x) - f(a)}{x - a} \right) (x - a) \right) = 0 \text{ [We have done this to use the fact that } f \text{ is differentiable].}$$

$$\text{Thus we can show that } \lim_{x \rightarrow a} \left(\left(\frac{f(x) - f(a)}{x - a} \right) (x - a) \right) = 0$$

Now since f is differentiable, $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists and $\lim_{x \rightarrow a} x - a$ exists because $x - a$ is continuous over $\mathbb{R}$, so $\lim_{x \rightarrow a} \left(\frac{f(x) - f(a)}{x - a} (x - a) \right) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} x - a = f'(a)(a - a) = f'(a)(0) = 0$ and thus f is continuous at a

We can also prove this by going backwards, i.e. Note that $f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a)$ and working with this to show $\lim_{x \rightarrow a} f(x) = f(a)$

Note that the converse is not true. Consider $f(x) = |x|$, then f is continuous at $x = 0$, but not differentiable

Proof that this is not differentiable:

$$f \text{ is differentiable at } x = a \Leftrightarrow \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists}$$

$$\text{Now } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h}, \text{ so at } x = 0, \text{ this becomes}$$

$$\lim_{h \rightarrow 0} \frac{|h|}{h}. \text{ Then } \lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = \lim_{h \rightarrow 0^+} (1) = 1 \text{ but } \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = \lim_{h \rightarrow 0^-} (-1) = -1, \text{ thus}$$

$$\lim_{h \rightarrow 0} \frac{|h|}{h} \text{ does not exist so } f'(0) \text{ does not exist}$$

$$\text{Note that } \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ since if we take } x = a + h, \text{ then as } x \rightarrow a, h \rightarrow 0$$

Theorem: Mean Value Theorem

Let f be a function such that:

- 1) f is continuous on the closed interval $[a, b]$
- 2) f is differentiable on the open interval (a, b)

Then, there exists a number $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$

Proof:

Consider the function h defined as: $h(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$ [$b \neq a$]

[Note this definition is such that $h(a) = h(b) = 0 \rightarrow$ Rolles]

f is continuous on $[a, b]$ and $(x-a)$ is continuous on $[a, b]$, thus h is continuous on $[a, b]$

[Note that $f(a)$ and $\frac{f(b)-f(a)}{b-a}$ are constants and don't affect continuity]

Since f and $(x-a)$ are differentiable on (a, b) , it follows that h is also differentiable on (a, b)

Now $h(a) = f(a) - f(a) - \frac{f(b)-f(a)}{b-a}(a-a) = 0$ and $h(b) = f(b) - f(a) - \frac{f(b)-f(a)}{b-a}(b-a) = 0$, thus, by Rolle's theorem, there exists a number $c \in (a, b)$ such that $h'(c) = 0$

Now $h'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$

Then, taking $x = c$, we get $h'(c) = f'(c) - \frac{f(b)-f(a)}{b-a} = 0 \Rightarrow f'(c) = \frac{f(b)-f(a)}{b-a}$

Theorem: Rolle's Theorem

Let f be a function such that:

- 1) f is continuous on a closed interval $[a, b]$
- 2) f is differentiable on an open interval (a, b)
- 3) $f(a) = f(b)$

Then, there exists a number $c \in (a, b)$ such that $f'(c) = 0$

Proof: The function f has two possible cases:

(i) Suppose f is a constant function on $[a, b]$.

Then $f'(x) = 0$ for all $x \in (a, b)$ so there exists some $c \in (a, b)$ where $f'(c) = 0$

(ii) Suppose f is not a constant function on $[a, b]$. Then $\exists$ a number $d \in (a, b)$ such that $f(d) \neq f(a)$. Now we either have $f(d) > f(a)$ or $f(d) < f(a)$ [giving 2 sub cases]

1. Suppose $f(d) > f(a)$. Now, by the Extreme Value Theorem, since f is continuous on $[a, b]$, there exists a number $c \in [a, b]$ such that $f(c)$ is an absolute maximum.

Thus $f(c) \geq f(d) > f(a)$, then, since $f(a) = f(b)$, $c \neq a$ and $c \neq b$ so $c \in (a, b)$, then since f is differentiable on (a, b) , we have that $f'(c)$ exists and thus, by Fermat's Theorem, $f'(c) = 0$

2. Suppose $f(d) < f(a)$. Now, by the Extreme Value Theorem, since f is continuous on $[a, b]$, there exists a number $c \in [a, b]$ such that $f(c)$ is an absolute minimum.

Thus $f(c) \leq f(d) < f(a)$. Now since $f(a) = f(b)$, we have that $c \neq a$ and $c \neq b$, thus $c \in (a, b)$, then since f is differentiable on (a, b) , we have that $f'(c)$ exists and thus, by Fermat's Theorem, $f'(c) = 0$

Note: for this proof, d is only used to prove that $c \neq a$ and $c \neq b$ to get the open interval (a, b) .

Theorem: Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

We do not prove this in MAM1000W

Note that since the absolute minimum and maximum values can occur at endpoints, there may not be an absolute min/max on an open interval (a, b)

One possible example is $f(x) = x$ on the interval $(1, 2)$. No such min/max exists

Also note that f must be continuous. If f is not, then there may not be a min/max

Theorem: Fermat's Theorem

If f has a local maximum (or minimum) at $c \in D_f$ **and** if $f'(c)$ exists, then $f'(c) = 0$

Note the converse is not true (for example, $f(x) = x^3$ has the property that $f'(0) = 0$ but this is not a local minimum or maximum)

Proof:

Suppose that f has a local maximum or minimum at $c \in D_f$ and $f'(c)$ exists.

This means there exists an open interval I such that $c \in I$, and for all $x \in I$, $f(c) \geq f(x)$.

Furthermore, $\lim_{h \rightarrow 0} \frac{f(c+h)-f(c)}{h}$ exists.

Now, since c is a local maximum, this means $f(c+h) \leq f(c) \Rightarrow f(c+h) - f(c) \leq 0$
 $\forall h$ such that $c+h \in D_f$. Then if $h \rightarrow 0^-$, then $h < 0$ and thus $\frac{f(c+h)-f(c)}{h} \geq 0$ meaning

$$\lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \geq \lim_{h \rightarrow 0^-} (0) = 0$$

Now if $h \rightarrow 0^+$, then $h > 0$ and thus $\frac{f(c+h)-f(c)}{h} \leq 0$ and so $\lim_{h \rightarrow 0^+} \frac{f(c+h)-f(c)}{h} \leq \lim_{h \rightarrow 0^+} (0) = 0$

Then since $\lim_{h \rightarrow 0} \frac{f(c+h)-f(c)}{h}$ exists, we have that $\lim_{h \rightarrow 0^-} \frac{f(c+h)-f(c)}{h} = \lim_{h \rightarrow 0^+} \frac{f(c+h)-f(c)}{h}$ and thus

$$0 \leq \lim_{h \rightarrow 0} \frac{f(c+h)-f(c)}{h} \leq 0, \text{ meaning } \lim_{h \rightarrow 0} \frac{f(c+h)-f(c)}{h} = 0 \Rightarrow f'(c) = 0$$

Theorem: If f is a differentiable function on (a, b) and $f'(x) = 0$ for all $x \in (a, b)$, then f is a constant function on (a, b)

Proof:

First note that f is constant $\Leftrightarrow \forall x_1, x_2 \in (a, b), f(x_1) = f(x_2)$ [definition of constant function]. Now suppose $x_1, x_2 \in (a, b)$ be any numbers and, without loss of generality, assume $x_1 < x_2$. We will show that $f(x_1) = f(x_2)$. Consider the closed interval $[x_1, x_2]$. Since f is continuous on (a, b) , it follows that f is continuous on $[x_1, x_2]$ (since $x_1, x_2 \in (a, b)$). Moreover, f is also differentiable on (x_1, x_2) , since f is differentiable (a, b) . Thus, by Mean Value Theorem, there exists a number $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2)-f(x_1)}{x_2-x_1} \Leftrightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1), \text{ then by the assumptions, } f'(c) = 0$$

and thus $f(x_2) = f(x_1)$

Corollary: If $f'(x) = g'(x)$ for all $x \in (a, b)$, then $f - g$ is a constant function on (a, b)

Proof:

Let $F(x) = f(x) - g(x)$. We will show that F is constant and thus $f - g$ is constant

Then $F'(x) = f'(x) - g'(x) = 0$, thus (by the above proof), F is a constant

Theorem: If f is differentiable on (a, b) , then:

(1) : If $f'(x) > 0$ for all $x \in (a, b)$, then f is increasing on (a, b)

(2) : If $f'(x) < 0$ for all $x \in (a, b)$, then f is decreasing on (a, b)

Proof:

Definition of increasing

(1): f is increasing on $(a, b) \Leftrightarrow$ for any $x_1, x_2 \in (a, b)$ such that $x_1 < x_2$, then $f(x_1) < f(x_2)$. Suppose $f'(x) > 0$ for all $x \in (a, b)$, and let $x_1, x_2 \in (a, b)$ with $x_1 < x_2$. Consider the closed interval $[x_1, x_2]$. f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) since f is differentiable on (a, b) so, by MVT, there exists $c \in (x_1, x_2)$ such that $f'(c)(x_2 - x_1) = f(x_2) - f(x_1)$, thus, since $f'(c) > 0$ and $x_2 > x_1$, we have that $f(x_2) > f(x_1)$

(2) follows similarly from this

Theorem: The triangle inequality, $|a + b| \leq |a| + |b|$

Proof:

This proof using the property: $|x| \leq a \Leftrightarrow -a \leq x \leq a$

First note that $-|b| \leq b \leq |b|$. Similarly, $-|a| \leq a \leq |a|$. Using both of these, we get that

$$-|a| - |b| \leq a + b \leq |a| + |b|$$

$$\Leftrightarrow -(|a| + |b|) \leq a + b \leq |a| + |b|$$

$$\Leftrightarrow |a + b| \leq |a| + |b|$$

Theorem: $|a - b| \geq |a| - |b|$

Proof:

We use the triangle inequality.

First note that $|a - b + b| = |a|$ and that $|a - b + b| \leq |a - b| + |b|$ so $|a| \leq |a - b| + |b|$

$$\Leftrightarrow |a| - |b| \leq |a - b|$$

Proving something like $|ab| = |a||b|$ we break this up into cases:

(i) $a \geq 0$ and $b \geq 0$. Then $|a| = a$ and $|b| = b$ so $|a||b| = ab$. Moreover, $ab \geq 0$ so $|ab| = ab = |a||b|$

(ii) $a \geq 0$ and $b < 0$. Then $|a| = a$ and $|b| = -b$ so $|a||b| = -ab$. Moreover, $ab < 0$ so $|ab| = -ab = |a||b|$

(iii) $a < 0$ and $b < 0$. Then $|a| = -a$ and $|b| = -b$ so $|a||b| = ab$. Moreover $ab \geq 0$ so $|ab| = ab = |a||b|$

(iv) $a < 0$ and $b \geq 0$. This case follows similarly to (ii)

Proof that $|a|^n = |a^n|$

We have 2 cases:

(i) $a \geq 0$. Then $|a| = a$ and thus $|a|^n = a^n$. Now $a^n \geq 0$ so $a^n = |a^n| = |a|^n$

(ii) $a < 0$. Then $|a| = -a$. Now $|a|^n = (-a)^n$. Since $a < 0$, $-a > 0$ so $(-a)^n > 0$ so $|(-a)^n| = |a|^n \Rightarrow |-a^n| = |a|^n \Leftrightarrow |a^n| = |a|^n$

Absolute values will almost always deal with cases

Theorem: The Factor Theorem

c is a root of $p(x) \Leftrightarrow x - c$ is a factor of $p(x)$.

Proof:

($\Rightarrow$) Suppose c is a root. Now consider division by $x - c$. Then $p(x) = q(x)(x - c) + r(x)$. We also have that $0 \leq \deg(r(x)) < \deg(x - c) \Leftrightarrow 0 \leq \deg(r(x)) < 1 \Leftrightarrow \deg(r(x)) = 0$ so $r(x) = a$ with $a \in \mathbb{R}$

Now $p(x) = q(x)(x - c) + a \Rightarrow p(c) = q(c)(c - c) + a \Leftrightarrow 0 = q(c)(0) + a \Leftrightarrow a = 0$ so $r(x) = 0$ so $x - c$ is a factor

($\Leftarrow$) Now suppose $x - c$ is a factor. Then $p(x) = q(x)(x - c) \Leftrightarrow p(c) = q(c)(c - c) \Leftrightarrow p(c) = 0$ so c is a root

Theorem: If $f(x) \leq g(x)$ on $(a - u, a + u) - \{a\}$ [some interval "close" to a except possibly at a] and both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$

We do not prove this

Theorem: Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ on $(a - u, a + u) - \{a\}$ and $\lim_{x \rightarrow a} f(x)$, $\lim_{x \rightarrow a} g(x)$, and $\lim_{x \rightarrow a} h(x)$ all exist and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$ then $\lim_{x \rightarrow a} g(x) = L$

Proof:

By the above theorem, $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x) \leq \lim_{x \rightarrow a} h(x) \Rightarrow L \leq \lim_{x \rightarrow a} g(x) \leq L \Rightarrow \lim_{x \rightarrow a} g(x) = L$

Note since $f(x) < g(x) \Rightarrow f(x) \leq g(x)$, we have that Squeeze theorem works for $f(x) < g(x) < h(x)$

Theorem: If for some function g and a real number a with $\lim_{x \rightarrow a} g(x) = b$ and f is continuous at b , then $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$

We do not prove this

Fact: $\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)}$ for $n \in \mathbb{Z}^+$

Proof: Let $f(x) = \sqrt[n]{x}$, then $f(g(x)) = \sqrt[n]{g(x)}$. Now if n is even, $f(x)$ is continuous over $[0, \infty)$ and thus $\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)}$ for $g(x) \geq 0$. If n is odd, $f(x)$ is continuous over $\mathbb{R}$ and thus $\lim_{x \rightarrow a} \sqrt[n]{g(x)} = \sqrt[n]{\lim_{x \rightarrow a} g(x)}$

We also use this to show that:

Fact: If g is continuous at a and f is continuous at $g(a)$, then $f(g(x))$ is continuous at a

Proof: Since g is continuous at a , $g(a) = \lim_{x \rightarrow a} g(x)$. Now f is continuous at $g(a) = b$, so by the above theorem: $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)) = f(g(a))$ and thus $(f \circ g)(x)$ is continuous at a

This is useful for proving continuity of compositions:

$h(x) = \cos(x^2)$. Since $h(x) = f(g(x))$ where $f(x) = \cos x$ and $g(x) = x^2$ and f is continuous over $\mathbb{R}$ and g is continuous over $\mathbb{R}$, h is continuous over $\mathbb{R}$

Lemma: For positive integers a, b , if $a > b$ then $a \nmid b$

Proof: Suppose $a > b$ and $a|b$ to obtain a contradiction. Then $\exists$ a positive integer k such that $b = ak$. Now $k \geq 1$, so $ak \geq a$. But then $b = ak \geq a$ [a is positive] so $b \geq a$ which is a contradiction and so $a \nmid b$

Lemma: For positive integers a, b such that $a > 1$. Then $a \nmid n$ or $a \nmid (n + 1)$

Proof: Suppose, to the contrary, that $a|n$ and $a|(n + 1)$. Then $\exists$ positive integers k, z such that $n = ak$ and $n + 1 = az$. So $ak + 1 = az \Rightarrow a(z - k) = 1$. Both $a, z - k \in \mathbb{Z}$ so $a = 1 = z - k$ or $a = -1 = z - k$ so $a \leq 1$ giving a contradiction. We can also prove it from $a(z - k) = 1 \Rightarrow a|1 \Rightarrow a \leq 1$. Or $a > 1$ means that $a \nmid 1$ which is a contradiction

Theorem: There are infinitely many prime numbers

Proof: Suppose there are finitely many primes $p_1, p_2, \dots, p_k$ for $k > 0$. Now consider $n = p_1 p_2 \dots p_k + 1$.

Then by the Fundamental Theorem of Arithmetic, n has its own unique prime factorisation and thus some $p_i | n$ with $1 \leq i \leq k$ for $i \in \mathbb{N}$. But now $n - 1 = p_1 p_2 \dots p_k$ and thus $p_i | (n - 1)$. This is a contradiction to the above Lemma and thus there are infinite primes

Lemma: Let p be a prime number and n a positive integer. Then $p | n \Leftrightarrow p | n^2$

Proof: First note that $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ by The Fundamental Theorem of Arithmetic. Then $n^2 = p_1^{2\alpha_1} p_2^{2\alpha_2} \dots p_k^{2\alpha_k}$ and thus n and n^2 have the same prime factors, so $p | n \Leftrightarrow p | n^2$

A conventional proof would go as follows: Suppose $p | n \Rightarrow n = pa \Rightarrow n^2 = p^2 a^2 = (pa^2)p \Rightarrow p | n^2$. Now suppose that $p \nmid n$ (contrapositive proof), so $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ with no $p_i = p$ for $1 \leq i \leq k$. Then $n^2 = p_1^{2\alpha_1} p_2^{2\alpha_2} \dots p_k^{2\alpha_k}$ and no $p_i = p$ for $1 \leq i \leq k$, so $p \nmid n^2$

Important Limits:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

This proof is done geometrically, or alternatively, use L'Hopital's Rule:

$$= \lim_{x \rightarrow 0} \frac{\cos(x)}{1} = \cos(0) = 1. \text{ See page on the geometric proof}$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = 0$$

Proof:

$$\lim_{x \rightarrow 0} \frac{(\cos(x)-1)(\cos(x)+1)}{x(\cos(x)+1)} = \lim_{x \rightarrow 0} \frac{\cos^2(x)-1}{x(\cos(x)+1)} = \lim_{h \rightarrow 0} \frac{-\sin^2(x)}{x(\cos(x)+1)} = \lim_{h \rightarrow 0} \frac{\sin(x)}{x} \cdot \lim_{h \rightarrow 0} \frac{-\sin(x)}{\cos(x)+1} = 1 \cdot 0 = 0$$

Differentiation Formulas

1. Constant Functions: *The derivative of a constant function is always 0*

Proof:

$$\text{Suppose } f(x) = c, \text{ then } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

2. The Constant Multiple Rule:

If f is a differentiable function and $c \in \mathbb{R}$, then $\frac{d}{dx}(cf(x)) = c \frac{d}{dx}(f(x))$

Proof:

$$\begin{aligned} \frac{d}{dx}(cf(x)) &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} = \lim_{h \rightarrow 0} \frac{c(f(x+h) - f(x))}{h} \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = cf'(x) \end{aligned}$$

3. The Sum Rule:

If f and g are differentiable functions, then $(f + g)'(x) = f'(x) + g'(x)$

Proof:

$$\begin{aligned} (f + g)'(x) &= \lim_{h \rightarrow 0} \frac{(f+g)(x+h) - (f+g)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) + g(x+h) - (f(x) + g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) + g(x+h) - g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) + g'(x) \end{aligned}$$

4. The Difference Rule:

If f and g are differentiable functions, then $(f - g)'(x) = f'(x) - g'(x)$

Proof:

$$\begin{aligned}(f - g)'(x) &= \lim_{h \rightarrow 0} \frac{(f - g)(x + h) - (f - g)(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h) - g(x + h) - (f(x) - g(x))}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x) - (g(x + h) - g(x))}{h} \\&= \lim_{h \rightarrow 0} \left[\frac{f(x + h) - f(x)}{h} - \frac{g(x + h) - g(x)}{h} \right] \\&= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} - \lim_{h \rightarrow 0} \frac{g(x + h) - g(x)}{h} \\&= f'(x) - g'(x)\end{aligned}$$

5. The Product Rule:

If f and g are differentiable functions, then

$$\frac{d}{dx}(f(x)g(x)) = f(x)\frac{d}{dx}(g(x)) + \frac{d}{dx}(f(x))g(x) \Leftrightarrow (fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

Proof:

$$\begin{aligned}(fg)'(x) &= \lim_{h \rightarrow 0} \frac{(fg)(x + h) - (fg)(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h)g(x + h) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h)g(x + h) - f(x)g(x) + f(x + h)g(x) - f(x + h)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h)g(x + h) - f(x + h)g(x) + f(x + h)g(x) - f(x)g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x + h)(g(x + h) - g(x)) + g(x)(f(x + h) - f(x))}{h} \\&= \lim_{h \rightarrow 0} \left[f(x + h) \frac{g(x + h) - g(x)}{h} \right] + \lim_{h \rightarrow 0} \left[g(x) \frac{f(x + h) - f(x)}{h} \right] \\&= f(x)g'(x) + g(x)f'(x)\end{aligned}$$

Since we know what $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ and $\lim_{h \rightarrow 0} \frac{g(x+h)-g(x)}{h}$, we need to factorise this into these forms. This means one of $f(x+h)$ or $g(x+h)$ must be added (to then factorise and give the other one of $f(x+h)$ or $g(x+h)$, thus the other term must match this and be either $f(x)$ or $g(x)$). This means if we subtract $f(x+h)g(x)$ we can factorise as $f(x+h)(g(x+h)-g(x))$ [or subtract $g(x+h)f(x)$ to give $g(x+h)(f(x+h)-f(x))$], thus either term can be added and subtracted, to keep the value of the expression the same. We will add and subtract $f(x+h)g(x)$.

6. The Quotient Rule:

If f and g are differentiable functions, then $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

Proof:

$$\begin{aligned} \left(\frac{f}{g} \right)'(x) &= \lim_{h \rightarrow 0} \frac{\left(\frac{f}{g} \right)(x+h) - \left(\frac{f}{g} \right)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)}}{h} \end{aligned}$$

Now by the same logic as above, $f(x)g(x)$ is added and subtracted

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{g(x)f(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x)g(x+h)} \\ &= \lim_{h \rightarrow 0} \frac{g(x)(f(x+h) - f(x)) - f(x)(g(x+h) - g(x))}{hg(x)g(x+h)} \\ &= \lim_{h \rightarrow 0} \left[\frac{1}{g(x+h)} \cdot \frac{f(x+h) - f(x)}{h} \right] - \lim_{h \rightarrow 0} \left[\frac{f(x)}{g(x)g(x+h)} \cdot \frac{g(x+h) - g(x)}{h} \right] \\ &= \frac{f'(x)}{g(x)} - \frac{f(x)f'(x)}{g(x)^2} = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \end{aligned}$$

7. The Chain Rule (Do not need to know)

Proof:

If f and g are differentiable functions, $(f \circ g)'(x) = f'(g(x))g'(x)$:

$$\begin{aligned}(f \circ g)'(x) &= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h} \\&= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \cdot \frac{g(x+h) - g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}\end{aligned}$$

Letting $g(x+h) = x' + h'$ and $g(x) = x'$, then as $h \rightarrow 0$, $g(x+h) \rightarrow g(x)$ and so $g(x+h) - g(x) \rightarrow 0$ so $x' + h' - x' \rightarrow 0$ meaning $h' \rightarrow 0$ and so:

$$\begin{aligned}(f \circ g)'(x) &= \lim_{h' \rightarrow 0} \frac{f(x' + h') - f(x')}{h'} \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = f'(x') \cdot g'(x) \\&= f'(g(x)) \cdot g'(x)\end{aligned}$$

Power Rule (General Case): $f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$

$y = x^n \Rightarrow |y| = |x^n|$, thus using logarithmic differentiation:

$$\ln|y| = \ln|x^n|$$

$$\Leftrightarrow \ln|y| = \ln|x|^n$$

$$\Leftrightarrow \ln|y| = n\ln|x|$$

$$\text{So } \frac{1}{y} y' = n \frac{1}{x} \Rightarrow y' = x^n \frac{n}{x} = nx^{n-1}$$

Trigonometric Derivatives

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

$$\begin{aligned}\frac{d}{dx}(\sin(x)) &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\sin(x)(\cos(h)-1) + \sin(h)\cos(x)}{h} \\&= \lim_{h \rightarrow 0} \left[\frac{\sin(x)(\cos(h)-1)}{h} + \frac{\sin(h)\cos(x)}{h} \right] \\&= \lim_{h \rightarrow 0} \frac{\sin(x)(\cos(h)-1)}{h} + \lim_{h \rightarrow 0} \frac{\sin(h)\cos(x)}{h} \\&= \lim_{h \rightarrow 0}(\sin(x)) \cdot \lim_{h \rightarrow 0} \frac{(\cos(h)-1)}{h} + \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \cdot \lim_{h \rightarrow 0}(\cos(x)) \\&= \sin(x) \cdot 0 + 1 \cdot \cos(x) = \cos(x)\end{aligned}$$

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

$$\begin{aligned}\frac{d}{dx}(\cos(x)) &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{\cos(x)(\cos(h)-1) - \sin(x)\sin(h)}{h} \\&= \lim_{h \rightarrow 0} \left[\frac{\cos(x)(\cos(h)-1)}{h} - \frac{\sin(x)\sin(h)}{h} \right] \\&= \lim_{h \rightarrow 0} \frac{\cos(x)(\cos(h)-1)}{h} - \lim_{h \rightarrow 0} \frac{\sin(x)\sin(h)}{h} \\&= \lim_{h \rightarrow 0} \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} - \lim_{h \rightarrow 0} \sin(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \\&= \cos(x) \cdot 0 - \sin(x) \cdot 1 = -\sin(x)\end{aligned}$$

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

$$\frac{d}{dx}(\tan(x)) = \frac{d}{dx}\left(\frac{\sin(x)}{\cos(x)}\right)$$

Then, by the quotient rule:

$$= \frac{\cos(x)\cos(x) - (-\sin(x))\sin(x)}{(\cos(x))^2}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)}$$

$$= \frac{1}{\cos^2(x)}$$

$$= \sec^2(x)$$

$$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

$$\frac{d}{dx}(\cot(x)) = \frac{d}{dx}\left(\frac{\cos(x)}{\sin(x)}\right)$$

Then, by the quotient rule:

$$= \frac{-\sin(x)\sin(x) - \cos(x)\cos(x)}{(\sin(x))^2}$$

$$= -\frac{\sin^2(x) + \cos^2(x)}{\sin^2(x)}$$

$$= -\frac{1}{\sin^2(x)} = -\csc^2(x)$$

$$\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$$

$$\text{Take } \sec(x) = \frac{1}{\cos(x)} = (\cos(x))^{-1}$$

$$\text{Thus } \frac{d}{dx}(\sec(x)) = \frac{d}{dx}((\cos(x))^{-1})$$

$$= -(\cos(x))^{-2} \cdot \frac{d}{dx}(\cos(x)) \text{ [by the Chain Rule]}$$

$$= -\frac{1}{(\cos(x))^2} \cdot -\sin(x) = \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{\cos(x)}$$

$$= \sec(x)\tan(x)$$

$$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$$

$$\begin{aligned}\frac{d}{dx}(\csc(x)) &= \frac{d}{dx}\left(\frac{1}{\sin(x)}\right) = \frac{d}{dx}((\sin(x))^{-1}) \\ &= -\sin(x)^{-2} \cdot \frac{d}{dx}(\sin(x)) \text{ [by the Chain Rule]} \\ &= -\frac{\cos(x)}{\sin(x)} \cdot \frac{1}{\sin(x)} \\ &= -\csc(x)\cot(x)\end{aligned}$$

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

Since $y = \ln(x) \Leftrightarrow x = e^y$, thus $\frac{d}{dx}(x) = \frac{d}{dy}(e^y)$

This gives $1 = e^y \cdot \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$

$$\frac{d}{dx}(\log_a(x)) = \frac{1}{x\ln(a)}$$

$\log_a x = \frac{\ln x}{\ln a}$, thus $\frac{d}{dx}(\log_a x) = \frac{1}{x\ln a}$

Inverse Derivative

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

This is true if f is a one-to-one function which is continuous and differentiable and if $f'(f^{-1}(a)) \neq 0$, then f^{-1} is differentiable

Proof:

$$(f^{-1})'(a) = \lim_{x \rightarrow a} \frac{f^{-1}(x) - f^{-1}(a)}{x - a}$$

Now $f^{-1}(a) \in D_f \Rightarrow$ there exists $b \in D_f, f^{-1}(a) = b$

Then $f(b) = a$, and claiming $f^{-1}(x) = y \Rightarrow f(y) = x$

Thus if $x \rightarrow a$, then $f^{-1}(x) \rightarrow f^{-1}(a) \Rightarrow y \rightarrow b$

$$\begin{aligned} (f^{-1})'(a) &= \lim_{x \rightarrow a} \frac{f^{-1}(x) - f^{-1}(a)}{x - a} = \lim_{y \rightarrow b} \frac{y - b}{f(y) - f(b)} \\ &= \lim_{y \rightarrow b} \frac{1}{\frac{f(y) - f(b)}{y - b}} = \frac{1}{\lim_{y \rightarrow b} \frac{f(y) - f(b)}{y - b}} = \frac{1}{f'(b)} = \frac{1}{f'(f^{-1}(a))} \end{aligned}$$

From this, we can see that if $f'(f^{-1}(a)) = 0$ then the equation is undefined.

TABLE OF RESULTS

Expressio n	Derivative	Expression (as a function of x)	Derivative
$\frac{d}{dx}(\sin(x))$	$\cos(x)$	$\frac{d}{dx}(\sin(y))$	$\cos(y) \cdot \frac{dy}{dx}$
$\frac{d}{dx}(\cos(x))$	$-\sin(x)$	$\frac{d}{dx}(\cos(y))$	$-\sin(y) \cdot \frac{dy}{dx}$
$\frac{d}{dx}(\tan(x))$	$\sec^2(x)$	$\frac{d}{dx}(\tan(y))$	$\sec^2(y) \cdot \frac{dy}{dx}$
$\frac{d}{dx}(\sec(x))$	$\sec(x)\tan(x)$	$\frac{d}{dx}(\sec(y))$	$\sec(y)\tan(y) \cdot \frac{dy}{dx}$
$\frac{d}{dx}(\cot(x))$	$-\csc^2(x)$	$\frac{d}{dx}(\cot(y))$	$-\csc^2(y) \cdot \frac{dy}{dx}$
$\frac{d}{dx}(\csc(x))$	$-\csc(x)\cot(x)$	$\frac{d}{dx}(\csc(y))$	$-\csc(y)\cot(y) \cdot \frac{dy}{dx}$

Derivative	Domain
$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$	$-1 < x < 1$
$(\arctan x)' = \frac{1}{1+x^2}$	$-\infty < x < \infty$
$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$	$-\infty < x < \infty$
$(\operatorname{arcsec} x)' = \frac{1}{ x \sqrt{x^2-1}}$	$x \in (-\infty, -1) \cup (1, \infty)$
$(\operatorname{arccsc} x)' = -\frac{1}{ x \sqrt{x^2-1}}$	$x \in (-\infty, -1) \cup (1, \infty)$